

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I **Bhuvaneshwari j** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

Given Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet using Resolution.

1. Propositional Logic :

Let:

- R = It is raining.
- W = The ground is wet.

Statement 1:

$R \rightarrow W$

Statement 2:

R

Goal: W

2. Convert into CNF:

Implication:

$R \rightarrow W$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore:

$\neg R \vee W$

Fact:

R

So the CNF clauses are:

- $C1: \neg R \vee W$
- $C2: R$

3. Negate the Goal

Goal:

W

Negated goal:

$\neg W$

Add:

- $C3: \neg W$

4. Resolution Steps

Resolve $C1$ and $C2$:

$C1: \neg R \vee W$

$C2: R$

Resolve $\neg R$ and R :

C4: W

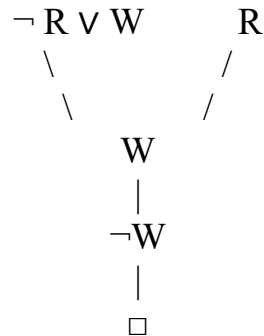
Now resolve C4 with C3:

C4: W

C3: $\neg W$

Therefore:

Resolution Tree :



Conclusion :

Since the Empty Clause ($\square$) is derived, the goal is proved.
Therefore, the ground is wet.

CODE :

Question 1: Rain and Wet Ground

$R \rightarrow W$

R is true

Goal: W

def resolve(clause1, clause2):

for literal in clause1:

opposite = literal[1:] if literal.startswith("~") else "~" + literal

if opposite in clause2:

return (clause1 - {literal}) | (clause2 - {opposite})

return None

CNF Clauses

C1 = {"~R", "W"} # $R \rightarrow W$

C2 = {"R"} # It is raining

C3 = {"~W"} # Negation of goal

print("Question 1: Rain and Wet Ground")

print("C1 =", C1)

print("C2 =", C2)

print("C3 =", C3)

Resolution Step 1

C4 = resolve(C1, C2)

print("\nResolution 1:")

print("C1 + C2 =", C4)

```
# Resolution Step 2
C5 = resolve(C4, C3)
print("\nResolution 2:")
print("C4 + C3 =", C5)
if C5 == set():
    print("\nEmpty Clause ( $\square$ ) obtained.")
    print("Conclusion: The ground is wet.")
```

OUTPUT:

```
Question 1: Rain and Wet Ground
C1 = {'W', '~R'}
C2 = {'R'}
C3 = {'~W'}

Resolution 1:
C1 + C2 = {'W'}

Resolution 2:
C4 + C3 = set()

Empty Clause ( $\square$ ) obtained.
Conclusion: The ground is wet.

=== Code Execution Successful ===
```

Question 2 – Student Assignment Submission :

Given Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

1. Propositional Logic :

Let:

- S = Rahul submitted the assignment.
- M = Rahul receives internal marks.

Statement 1:

$S \rightarrow M$

Statement 2:

S

Goal : M

2. Convert into CNF:

Statement 1:

$S \rightarrow M$

Therefore:

$\neg S \vee M$

Statement 2:

S

CNF clauses:

- C1: $\neg S \vee M$
- C2: S

3. Negate the Goal :

Goal:

M

Negated goal:

$\neg M$

Add:

- C3: $\neg M$

4. Resolution Steps :

Resolve C1 and C2:

C1: $\neg S \vee M$

C2: S

C4: M

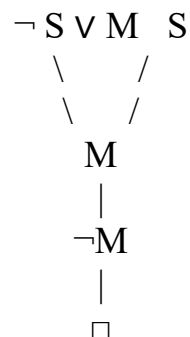
Now resolve C4 with the negated goal:

C4: M

C3: $\neg M$

□

Resolution Tree :



Conclusion :

The Empty Clause $\square$ is obtained.

Therefore, Rahul receives internal marks.

CODE :

```
# Question 2: Student Assignment Submission
# S -> M
# S is true
# Goal: M
def resolve(clause1, clause2):
    for literal in clause1:
        opposite = literal[1:] if literal.startswith("~") else "~" + literal
        if opposite in clause2:
            return (clause1 - {literal}) | (clause2 - {opposite})
    return None
# CNF Clauses
C1 = {"~S", "M"} # S -> M
C2 = {"S"}       # Rahul submitted assignment
C3 = {"~M"}      # Negation of goal
print("Question 2: Student Assignment Submission")
print("C1 =", C1)
print("C2 =", C2)
print("C3 =", C3)
# Resolution Step 1
C4 = resolve(C1, C2)
print("\nResolution 1:")
print("C1 + C2 =", C4)
# Resolution Step 2
C5 = resolve(C4, C3)
print("\nResolution 2:")
print("C4 + C3 =", C5)
if C5 == set():
    print("\nEmpty Clause ( $\square$ ) obtained.")
    print("Conclusion: Rahul receives internal marks.")
```

OUTPUT:

```
Question 2: Student Assignment Submission
```

```
C1 = {'M', '~S'}
```

```
C2 = {'S'}
```

```
C3 = {'~M'}
```

```
Resolution 1:
```

```
C1 + C2 = {'M'}
```

```
Resolution 2:
```

```
C4 + C3 = set()
```

```
Empty Clause ( $\square$ ) obtained.
```

```
Conclusion: Rahul receives internal marks.
```

```
=== Code Execution Successful ===
```

Question 3 – Library Membership :

Given Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books.

1. Propositional Logic :

Let:

- L = Priya is a library member.
- B = Priya can borrow books.

Statement 1:

$L \rightarrow B$

Statement 2:

L

Goal:

B

2. Convert into CNF :

Statement 1:

$L \rightarrow B$

Therefore:

$\neg L \vee B$

Statement 2:

L

CNF clauses:

- C1: $\neg L \vee B$
- C2: L

3. Negate the Goal :

Goal:

B

Negated goal:

$\neg B$

Add:

- C3: $\neg B$

4. Resolution :

Resolve C1 and C2:

C1: $\neg L \vee B$

C2: L

C4: B

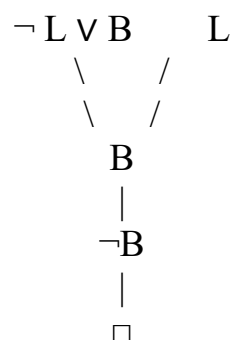
Now resolve C4 with C3:

C4: B

C3: $\neg B$

□

Resolution Tree :



Conclusion :

- The Empty Clause □ is derived.
- Therefore, Priya can borrow books.
- The goal is proved by resolution.

CODE :

Question 3: Library Membership

L $\rightarrow$ B

L is true

Goal: B

```

def resolve(clause1, clause2):
    for literal in clause1:
        opposite = literal[1:] if literal.startswith("~") else "~" + literal
        if opposite in clause2:
            return (clause1 - {literal}) | (clause2 - {opposite})
    return None

# CNF Clauses
C1 = {"~L", "B"} # L -> B
C2 = {"L"} # Priya is a library member
C3 = {"~B"} # Negation of goal
print("Question 3: Library Membership")
print("C1 =", C1)
print("C2 =", C2)
print("C3 =", C3)
# Resolution Step 1
C4 = resolve(C1, C2)
print("\nResolution 1:")
print("C1 + C2 =", C4)
# Resolution Step 2
C5 = resolve(C4, C3)
print("\nResolution 2:")
print("C4 + C3 =", C5)
if C5 == set():
    print("\nEmpty Clause ( $\square$ ) obtained.")
    print("Conclusion: Priya can borrow books.")

```

OUTPUT:

```

Output
Question 3: Library Membership
C1 = {'~L', 'B'}
C2 = {'L'}
C3 = {'~B'}

Resolution 1:
C1 + C2 = {'B'}

Resolution 2:
C4 + C3 = set()

Empty Clause ( $\square$ ) obtained.
Conclusion: Priya can borrow books.

=== Code Execution Successful ===

```

Question 4 – Placement Eligibility :

Given Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement.

1. Propositional Logic :

Let:

- A = Arun cleared the aptitude test.
- E = Arun is eligible for placement.

Statement 1:

$A \rightarrow E$

Statement 2:

A

Goal:

E

2. Convert to CNF :

Statement 1:

$A \rightarrow E$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore:

$\neg A \vee E$

Statement 2:

A

CNF clauses:

- $C1: \neg A \vee E$
- $C2: A$

3. Negate the Goal :

Goal:

E

Negated goal:

$\neg E$

Add:

- $C3: \neg E$

4. Resolution Steps :

First resolve $C1$ and $C2$:

$C1: \neg A \vee E$

$C2: A$

C4: E

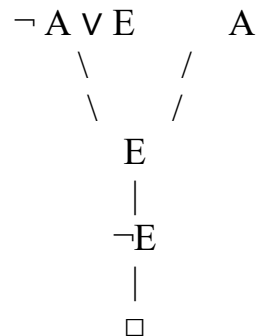
Then resolve C4 and C3:

C4: E

C3: $\neg E$

□

Resolution Tree :



Conclusion :

Since the Empty Clause (□) is obtained, the negation of the goal leads to a contradiction.

Therefore:

Arun is eligible for placement.

Hence, the goal is proved using Resolution.

CODE :

```
# Question 4: Placement Eligibility
# A -> E
# A is true
# Goal: E
def resolve(clause1, clause2):
    for literal in clause1:
        opposite = literal[1:] if literal.startswith("~") else "~" + literal
        if opposite in clause2:
            return (clause1 - {literal}) | (clause2 - {opposite})
    return None
# CNF Clauses
C1 = {"~A", "E"} # A -> E
C2 = {"A"} # Arun cleared aptitude test
C3 = {"~E"} # Negation of goal
print("Question 4: Placement Eligibility")
print("C1 =", C1)
print("C2 =", C2)
print("C3 =", C3)
# Resolution Step 1
```

```

C4 = resolve(C1, C2)
print("\nResolution 1:")
print("C1 + C2 =", C4)
# Resolution Step 2
C5 = resolve(C4, C3)
print("\nResolution 2:")
print("C4 + C3 =", C5)
if C5 == set():
    print("\nEmpty Clause ( $\square$ ) obtained.")
    print("Conclusion: Arun is eligible for placement.")

```

OUTPUT:

```

} Question 4: Placement Eligibility
C1 = {'E', '~A'}
C2 = {'A'}
C3 = {'~E'}

Resolution 1:
C1 + C2 = {'E'}

Resolution 2:
C4 + C3 = set()

Empty Clause ( $\square$ ) obtained.
Conclusion: Arun is eligible for placement.

=== Code Execution Successful ===

```

Question 5 – Access Control System :

Given Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access.

1. Propositional Logic :

Let:

- P = User entered the correct password.
- A = User is authenticated.

- G = User is granted access.

Statements

Statement 1:

$P \rightarrow A$

Statement 2:

$A \rightarrow G$

Statement 3:

P

Goal:

G

2. Convert All Implications into CNF :

Statement 1

$P \rightarrow A$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore:

$\neg P \vee A$

Statement 2

$A \rightarrow G$

Therefore:

$\neg A \vee G$

Statement 3

P

Therefore, CNF clauses are:

- $C1: \neg P \vee A$
- $C2: \neg A \vee G$
- $C3: P$

3. Negate the Goal :

Goal:

G

Negated goal:

$\neg G$

Add:

- $C4: \neg G$

4. Apply Resolution Algorithm :

Step 1

Resolve $C1$ and $C3$.

$C1: \neg P \vee A$

$C3: P$

$C5: A$

Because $\neg P$ and P cancel.

So:
C5 = A

Step 2
Resolve C2 and C5.

C2: $\neg A \vee G$

C5: A

C6: G

Because $\neg A$ and A cancel.

So:

C6 = G

Step 3
Resolve C6 with the negated goal C4.

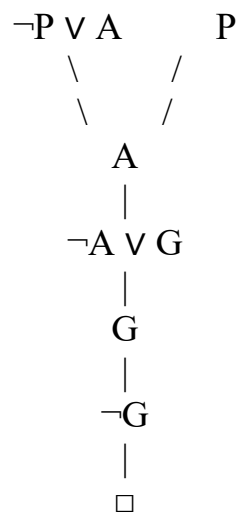
C6: G

C4: $\neg G$

 $\square$

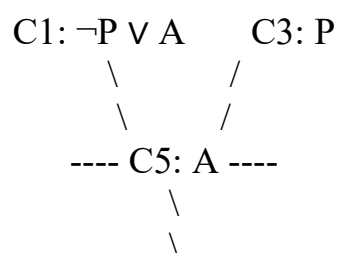
Therefore, we obtain the Empty Clause ($\square$).

Resolution Tree :



(Or)

showing all clauses:



$$\begin{array}{c}
 \text{C2: } \neg A \vee G \\
 | \\
 \text{C6: } G \\
 | \\
 \text{C4: } \neg G \\
 | \\
 \square
 \end{array}$$

General pattern:

$$P \rightarrow Q \quad \rightarrow \quad \neg P \vee Q$$

$$P \quad \quad P$$

$$\text{Goal: } Q \quad \neg Q \leftarrow \text{Negated Goal}$$

Resolution:

$$\neg P \vee Q + P \rightarrow Q$$

$$Q + \neg Q \rightarrow \square$$

Therefore, Q is proved.

CODE :

```

# Question 5: Access Control System
# P -> A
# A -> G
# P is true
# Goal: G
def resolve(clause1, clause2):
    for literal in clause1:
        opposite = literal[1:] if literal.startswith("~") else "~" + literal
        if opposite in clause2:
            return (clause1 - {literal}) | (clause2 - {opposite})
    return None
# CNF Clauses
C1 = {"~P", "A"} # P -> A
C2 = {"~A", "G"} # A -> G
C3 = {"P"} # Correct password entered
C4 = {"~G"} # Negation of goal
print("Question 5: Access Control System")
print("C1 =", C1)
print("C2 =", C2)
print("C3 =", C3)
print("C4 =", C4)
# Resolution Step 1
C5 = resolve(C1, C3)

```

```
print("\nResolution 1:")
print("C1 + C3 =", C5)
# Resolution Step 2
C6 = resolve(C2, C5)
print("\nResolution 2:")
print("C2 + C5 =", C6)
# Resolution Step 3
C7 = resolve(C6, C4)
print("\nResolution 3:")
print("C6 + C4 =", C7)
if C7 == set():
    print("\nEmpty Clause ( $\square$ ) obtained.")
    print("Conclusion: The user is granted access.")
```

OUTPUT:

```
Question 5: Access Control System
C1 = {'~P', 'A'}
C2 = {'G', '~A'}
C3 = {'P'}
C4 = {'~G'}

Resolution 1:
C1 + C3 = {'A'}

Resolution 2:
C2 + C5 = {'G'}

Resolution 3:
C6 + C4 = set()

Empty Clause ( $\square$ ) obtained.
Conclusion: The user is granted access.

=== Code Execution Successful ===
```